FARDIN IMRAN SHAIKH

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PROFESSIONAL SUMMARY

IoT Development Engineer with 2+ years of experience designing end-to-end IoT solutions with data collection and visualization capabilities. Skilled in sensor integration, IoT protocols, cloud platforms, real-time data processing, and actionable insights generation. Proven track record of building scalable data-driven applications that transform raw sensor data into business intelligence through advanced analytics, visualization dashboards, and predictive modelling.

TECHNICAL SKILLS

Programming & Scripting: Python, SQL, C/C++ (Arduino), HTML/CSS/JavaScript
Data Visualization & Analytics: Power BI, Excel (Pivot Tables, Charts, Basic Analysis), Real-Time IoT Dashboards, Data Cleaning, Basic Statistical Analysis
Database & Storage: MySQL, SQL Queries, Cloud Data Storage
IoT Platforms & Cloud: Raspberry Pi, Arduino, ESP32, ThingSpeak, ThingsBoard, Node-RED, Adafruit IO, Blynk IoT, Grafana
IoT Protocols: MQTT, HTTP, SPI, UART, I2C, LoRa
AI & Computer Vision: YOLOv8/YOLOv10 (Object Detection), Basic Machine Learning Concepts
Tools & Technologies: Git/GitHub, VS Code, Jupyter Notebook, VNC, Raspberry Pi Connect, Figma, 3D Modelling Prototyping & Printing

PROFESSIONAL EXPERIENCE

IoT Development Engineer & Application Trainer

Grok Learning Pvt Ltd, Mumbai, Maharashtra | April 2024–Present

- Developed and deployed end-to-end industry-levelIoT applications with integrated data collection and monitoring systems for smart agriculture, environmental monitoring, and automation use cases
- Designed real-time data pipelines using MQTT, Node-RED, and cloud platforms (ThingSpeak, ThingsBoard) to collect, store, and visualize sensor data with minimal latency
- Created interactive dashboards and data visualization solutions using Power BI and Python libraries to present IoT sensor data in actionable formats for monitoring and decision-making
- Performed data analysis on IoT sensor datasets using Python to identify trends, patterns, and system optimization opportunities
- Integrated SQL databases with IoT systems for historical data storage, enabling data querying and basic reporting for monitoring applications
- Coded and interfaced sensors with microcontrollers (ESP32, Raspberry Pi, Arduino) using Python and C++, ensuring reliable data acquisition and communication
- Implemented IoT communication protocols including MQTT, HTTP, and LoRa for device-to-device and device-to-cloud connectivity
- Performed comprehensive testing and debugging of IoT systems, troubleshooting hardware, software, and connectivity issues to ensure 99%+ reliability
- Conducted technical training sessions on IoT concepts, Python programming, data visualization basics, and 3D prototyping, successfully training 1500+ participants

Robotics Trainer

Atal Innovation Mission (NITI Aayog - Hope Foundation), Mumbai | June 2023 – March 2024

- Designed and delivered engaging robotics workshops and training sessions at Atal Tinkering Labs, inspiring students to explore STEM fields
- Mentored students in conceptualizing, designing, and building hands-on robotics projects using Arduino, Raspberry Pi, and other microcontrollers

- Developed curriculum integrating robotics programming, electronics, sensor data collection, and basic data logging into educational frameworks
- Taught technical skills in robotics programming, circuit design, and problem-solving to 500+ students, including basic data collection from sensors
- Successfully guided student teams in completing 15+ robotics projects from concept to working prototype with data monitoring features

KEY PROJECTS

AgriConnect: Smart Farming IoT Solution | Python, MQTT, SQL, Dashboard Visualization

- Built end-to-end IoT platform for precision agriculture with real-time soil monitoring (moisture, temperature, pH) and automated irrigation control
- Developed data collection pipeline using MQTT to store multi-sensor data in MySQL database for historical tracking and analysis
- Created real-time monitoring dashboard displaying sensor readings, trends, and automated alerts for irrigation optimization

SmartAssist Vision: AI-Powered Mobility Aid | Python, YOLOv8, IoT, Real-Time Processing

- Engineered IoT-enabled mobility assistance device using computer vision (YOLOv8) for real-time object detection and obstacle avoidance
- Implemented data logging system to track detection patterns and device usage for system improvement and testing

NestHawk: AI-Powered Autonomous Bird Detection & Pool Protection System | Python, YOLOv8-TFLite, Raspberry Pi, Telegram API, GPIO, IoT

- Engineered a fully autonomous smart bird detection system using YOLOv8-TFLite running on Raspberry Pi 4 with 95%+ detection accuracy at under 200ms response time
- Converted YOLOv8 model from PyTorch (.pt) to TensorFlow Lite INT8 quantized format (.tflite) for optimized edge inference on embedded hardware
- Integrated GPIO relay control to trigger precision sprinkler deterrence within 200ms of bird detection, with a 5-second burst and 30-second cooldown mechanism to prevent false triggers
- Implemented real-time Telegram Bot API alerts with photo evidence capture, delivering instant notifications to users upon bird detection events

EDUCATION

Bachelor of Science in Information Technology | CGPA: 7.79/10

Smt. K.G. Mittal Institute of Management, Information Technology & Research, Mumbai | 2020 – 2023

Relevant Coursework: Data Structures, Database Management Systems, Python Programming, Statistics

Higher Secondary Certificate (HSC) in Science - Electronics

Shri T.P. Bhatia Jr. College of Science | Completed March 2019

KEY ACHIEVEMENTS

- Developed IoT applications with data collection, storage, and visualization capabilities for real-world problem-solving
- Successfully trained 1500+ Participants in IoT development, Python programming, and basic data visualization across various workshops
- Proficient in designing end-to-end solutions combining hardware, software, and data visualization for scalable smart systems